

FINAL SESSION | REPLACES CATCH-UP SESSIONS 1-4

Finish the table in 1 class

Due End of class Tuesday. One block, about 60 minutes.

You are at Herringbone (3/4" thick) glued across the whole plywood (a bit under 19 1/2", one edge not straight), sanded, with a few small gaps. Base painted white, some screw holes show. Border mitered and dry-fit, with small gaps at the corners. No edge bands, no cleats yet, and the shelf is a rough blank.

Today's goal A table that looks finished: the top trimmed to fit inside its walnut border, screwed to the base, one clear coat, 4 photos. Bands, shelf, second coat, and screw holes come later (page 4).

Say this to your teacher in the first 3 minutes

"I was out sick Tuesday and Wednesday and gone Friday, so I rewrote my plan to finish in one class. Here it is. My border goes around the herringbone now, so it covers the plywood edge and I don't have the edge bands yet. Is there anything you need to see today for full credit? And can I get make-up time for the shelf, the bands, and a second coat?"

Only your teacher decides the grade. A written plan plus asking first is the best way to protect it, and teachers often give make-up time for sick days. Also ask them to watch your first table saw cut.

The 1-hour plan

Minutes	Task	Move on when
0-3	Show your teacher this page	Teacher OKs the plan
3-9	Fix the border frame; measure its opening; fill the herringbone gaps	Opposite border pieces match; opening written down
9-17	Trace the cut lines; trim the top to fit the border (4 saw cuts, crooked side first)	Border fits around it with the miters closed
17-24	Glue and clamp the border, face down	All 4 miters closed; face flat on the bench
24-34	Cut, drill, and install 4 top cleats	Flush with the apron top; 8 screws snug
34-38	Clamp off; scrape glue; sand filler and edges	Face smooth; no glue showing
38-44	Screw the top on	Same overhang on all 4 sides (about 3/4")
44-47	Check rocking; take 4 photos	All 4 feet touch; 4 photos saved
47-53	Clear coat on the top	Thin and even, no runs
53-60	Clean up; turn in; hand in page 4	Teacher knows the coat is wet

If you fall behind

- Never skip** trimming the top, gluing on the border, or the 4 photos.
- Skip first** the clear coat. It is the quickest thing to add next time.
- Skip second** screwing the top on. Set the top on the base for the photos and tell your teacher the screws go in next time.
- Checkpoints** Border glued by minute 24: if not, plan on rule 2. Cleats in by minute 34: if not, use rule 3 as well.

Bring

Herringbone top, 4 border pieces, base, 3/4" walnut or hardwood scrap for 4 cleats, 16 screws #8 x 1 1/4", glue, wax paper, band clamp, 2 clamps, walnut wood filler (or glue plus walnut sanding dust), putty knife, pin or brad nailer with 2" fasteners if the shop has one, 180 paper and a block, scraper, damp rag, clear satin water-based poly, brush, clean blanket, drill, 11/64" 3/16" 7/64" bits, countersink, painter's tape, square, tape measure, pencil, phone, these pages.

PART 1 OF 2 | MINUTES 0-24

Trim the top and glue on the border

Finish line The top is one piece, about 19 1/2" square: the herringbone trimmed to fit inside the border, the miters closed, and the face flush all the way around.

Why trim The herringbone runs to the plywood edges, so there is no room for the border on top. Your border is already mitered, so it sets the size: the herringbone gets cut to fit its inside opening.

Build steps

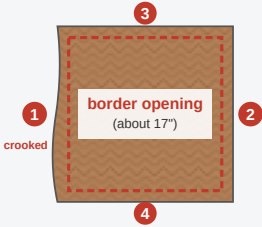
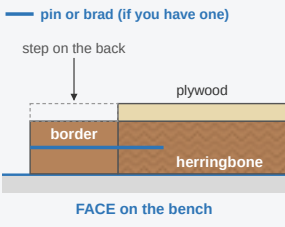
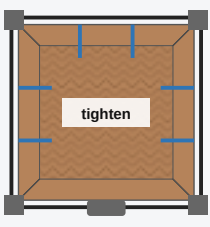
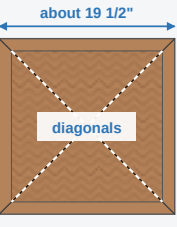
- 1 Check in** Show your teacher page 1 and ask the question in the box.
- 2 Fix the border frame** Lay the 4 border pieces in a frame on a flat bench. Put opposite pieces back to back: their long points should match. If one is longer, shave a hair off one end at the miter saw (still at 45°) until they match. Hairline corner gaps will close in the band clamp. Then measure the inside opening both ways: _____ x _____ (about 17"). Border width _____ thickness _____
- 3 Fill the gaps** Press walnut filler into the herringbone gaps with a putty knife and scrape it flat. No filler? Mix a few drops of glue with fine walnut dust from the sander bag. You sand it in step 12.
- 4 Trace the cut lines** Set the border frame on top of the herringbone, lined up with the straight edges. Trace along the inside of the frame on all 4 sides. Lift it off and put an X outside each line; that's the waste. The crooked side ends up in the waste.
- 5 Trim to fit** Table saw, herringbone face up (picture 1). **Cut 1:** the crooked side, with the straight side against the fence; slide the fence until the blade lines up just outside line 1, on the X side. **Cut 2:** cut-1 edge on the fence, fence at the opening (A or B) plus 1/32"; the blade should land on the X side of the line. **Cut 3:** crosscut sled, cut-1 edge against the sled fence, line 3 right next to the slot on the X side. **Cut 4:** cut-3 edge on the fence, fence at the other opening number plus 1/32". Test the frame around it. Too tight? Move the fence 1/32" and shave that side.
- 6 Dry fit face down** Wax paper on a flat bench. Top **face down**, border pieces face down around it, miters closed and tight to the herringbone. Face down is what makes the face flush (picture 2).
- 7 Glue and clamp** Glue the herringbone's edges and the miters, but keep glue off the 1/4" nearest the bench so none squeezes onto the face. Band clamp around the border; tighten until all 4 miters close (picture 3). Pin or brad nailer? Two 2" fasteners per side through the border into the herringbone, 3/8" up from the bench, and one across each corner. No nailer? The band clamp alone is fine. Leave it on until minute 34.
- 8 Check before it sets** Both diagonals match within 1/16" and the face is flat on the wax paper (picture 4). Wipe any glue you can reach with a damp rag.

Stop point at minute 24

[] The top is one piece, miters closed, and the band clamp is on.

If a machine is busy Fill gaps, measure, or cut the 4 cleats (step 9) while you wait.

Picture check

1 Cut order  Cut 1 is the crooked side, with the straight side on the fence. Cuts 2 and 4 on the fence, cut 3 on the sled.	2 Face down = flush face  The border is thinner than the top, so glue face down. The face comes out flush and the step ends up underneath.	3 Band clamp  Tighten until all 4 miters close. Blue = pins or brads, if you have a nailer: 2 per side, plus 1 across each corner.	4 Check before it sets  Measure corner to corner both ways. Equal means square. Face flat, miters closed, border tight to the herringbone.
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PART 2 OF 2 | MINUTES 24-60

Cleats, assemble, photograph, coat

Finish line The top is screwed on with the same overhang on every side, the table does not rock, 4 photos are saved, and the top has one clear coat.

Only 4 cleats today The 4 top cleats hold the top. The 4 shelf cleats go in with the shelf later.

Build steps

- 9 Make 4 top cleats** Rip a $\frac{3}{4}$ " x $\frac{3}{4}$ " strip at least 56" long from straight scrap and cut 4 pieces at 13 $\frac{1}{2}$ ". Check one between two legs inside an apron: you should see about $\frac{3}{4}$ " to spare at each end (leg to leg is 15").
- 10 Drill the cleats** At the drill press (picture 1). Side face: $\frac{11}{64}$ " holes at 1 $\frac{1}{2}$ " and 12", no countersink. Bottom face: $\frac{3}{16}$ " holes at 3 $\frac{3}{4}$ " and 9 $\frac{3}{4}$ ", shallow countersink with the head just flush.
- 11 Install the cleats** One inside each apron, $\frac{3}{4}$ " from each leg, top flush with the apron top, countersinks facing **down** (picture 2). Clamp it, drill a $\frac{7}{64}$ " pilot through each side hole with tape 1 $\frac{1}{4}$ " from the tip, and drive 1 $\frac{1}{4}$ " screws. Do not scratch the paint.
- 12 Clean up the top** Take off the band clamp and flip the top face up. Scrape glue at the joints and wipe with a damp rag. Sand the filled gaps flush with 180 on a block, then make 2 light passes on the outside edges of the border so they are not sharp. Carry the top by the herringbone, not the border, for the rest of class.
- 13 Screw the top on** Top face down on a clean blanket. Set the base upside down on it and measure until the overhang is the same on all 4 sides, about $\frac{3}{4}$ " (picture 3). Tape the $\frac{7}{64}$ " bit 1 $\frac{1}{4}$ " from the tip, drill through the 8 cleat holes until the tape touches, and drive the screws snug without burying the heads.
- 14 Flip and check** Upright on the flattest bench. All 4 feet should touch. Never shorten a leg without your teacher.
- 15 Take 4 photos** Front, X side, top from above, and a close-up of the pattern and a half-lap. Take them before the coat (picture 4).
- 16 Clear coat** Stir, do not shake. One thin coat on the face, then the border's outside edges, every stroke going the same direction. Catch drips under the edges. Keep it off the white paint.
- 17 Turn it in** Tell your teacher the coat is wet. Hand in page 4 as your record of changes.

Stop point at minute 60

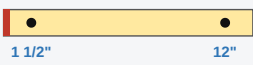
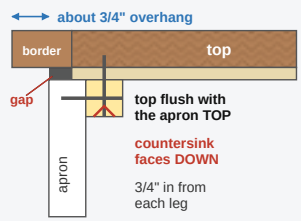
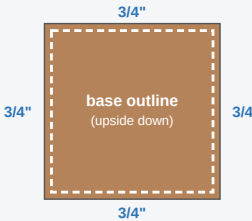
[] The table does not rock, and the overhang is the same on every side.

[] All 4 photos are saved.

[] The top has one coat, your teacher knows it is wet, and page 4 is handed in.

If you run late Skip the coat first, then the top screws (page 1). The photos always happen.

Picture check

1 Drill 4 cleats SIDE: $\frac{11}{64}$", no countersink  1 $\frac{1}{2}$" 12" BOTTOM: $\frac{3}{16}$", countersink  3 $\frac{3}{4}$" 9 $\frac{3}{4}$" END 13 $\frac{1}{2}$" long Measure every hole from the red END, centered across the cleat. 2 side holes instead of 3 saves time and still holds.	2 Where the top cleats go  about $\frac{3}{4}$" overhang border top gap apron top flush with the apron TOP countersink faces DOWN $\frac{3}{4}$" in from each leg Screws go into the apron and up into the herringbone. The small gap under the border gets fixed later (page 4).	3 Center the top  $\frac{3}{4}$" $\frac{3}{4}$" base outline (upside down) $\frac{3}{4}$" Top face down on a clean blanket. Make all 4 overhangs match before drilling.	4 4 photos ☐ 1 Front ☐ 2 X side ☐ 3 Top from above ☐ 4 Pattern + half-lap Take them before the coat. Wet water-based finish looks milky until it dries.
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HAND THIS IN | RECORD OF CHANGES

What changed, and what comes later

Absent _____ (sick), _____ (sick), _____ Periods used _____ of 10 planned

Record of changes

Change	Original packet	What I did	Why
Top layers	3/4" plywood + 1/4" walnut	1/4" plywood + 3/4" walnut	Material change. The top is still 1" thick
Build order	Border glued on the plywood, then herringbone inside it	Herringbone glued across the whole plywood, then trimmed to fit inside the border (opening _____ x _____)	Out 3 days. The pattern and the walnut frame look the same
Border	1 1/2" wide, 20" long point, sits on the plywood	Mitered, about 19 1/2" long point, _____ wide x _____ thick, glued around the herringbone	The border now sets the size of the top
Top glue-up	Clamps; glue sets 5 hours	Glued face down in a band clamp	One class left. Face down keeps the face flush
Edge bands	1/4" x 1" walnut around the outside	Not added yet	The border now covers the plywood edge. Can be added later
Top size	20" square, 1" overhang	About 19 1/2" square, about 3/4" overhang	No bands yet
Cleats	8 cleats, 3 side screws each	4 top cleats with 2 side screws each; 4 shelf cleats later	Time. Two screws per cleat hold the top
Shelf S1	14 7/8" square	Rail to rail minus 1/8" (about 15 5/8"), notched corners; added later	Rails sit 3/8" outside the legs, so the old shelf's screws missed it
Finish	2 coats before assembly	1 coat after assembly; second coat later	Never handle wet parts

Screws 16 today (8 cleat-to-apron, 8 top). 16 more later for the shelf. All #8 x 1 1/4".

Come back later, in this order

- 1 Clear coat** If you skipped it today, coat the top first.
- 2 Take the top off** Unscrew the 8 top screws. The shelf has to drop in from above, and the next two jobs are easier with the top off.
- 3 Close the gap under the border** Without bands, there is a thin gap between the border and the apron top that shows from the side. Pick one fix. **Bands** (if your teacher wants them): sand the finish off the border's outside edges with 120, then glue and pin four mitered 1/4" x 1" walnut bands around the top, bottoms flush with the plywood. The top becomes about 20" with a 1" overhang. **Filler strips** (no bands): measure the step under the border, rip strips of plywood that thick and about 3/4" wide, and glue them into the step tight against the plywood edge. Then give the underside one thin coat.
- 4 Shelf cleats** Make 4 more cleats, drilled exactly like the top cleats. Install them inside the rails, bottoms flush with the rail bottoms, countersinks down.
- 5 Shelf** Measure rail to rail both ways and subtract 1/8" (about 15 5/8"). Cut your blank square with the grain front to back and mark FRONT. Bandsaw a notch at each corner, about 3/8" x 3/8", so it clears the legs (old catch-up page 2, Fix 2). Sand 120 then 180, ease the edges, coat it, and let it dry.
- 6 Install the shelf** Base upright, shelf on the cleats with FRONT forward and 1/16" spacers in the gaps, clamp it. Flip the base upside down on a clean blanket and screw up through the 8 cleat holes with the taped 7/64" bit. Screw the top back on through its same 8 holes.
- 7 Holes** Fill the visible base screw holes, sand them flush, and touch up the white. Fill any pin or brad holes in the border with walnut filler.
- 8 Second coat** 220 grit with almost no pressure, wipe with a damp towel, then one thin coat on the top and the shelf. Retake the 4 photos if your teacher wants them with the shelf in.